METHANOL CONVERTER VER 1

Operator Training Simulator · R-201 quench converter and synthesis loop

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ELECTRICAL LOAD LIST

R-201 synthesis loop · C-201 drive, auxiliaries, UPS, heat tracing, lighting · cable sizing example

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DOCUMENT TITLE : Electrical Load List

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Teaching document generated from the Methanol Converter Ver 1 simulator model; numbers come from the model and the OTS tabs. Not for construction.

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1 Purpose and scope

This electrical load list gives every electrical consumer of the Methanol Converter Ver 1 synthesis loop (R-201, C-201, E-201 … E-204, V-201 and their instruments) with its rated and absorbed power, efficiency, power factor, duty and demand factor, and the resulting maximum demand on the 11 kV and 400 V systems. It also records the sizing of the C-201 motor and its feeder cable. Only the C-201 absorbed power comes from the simulator model; all other figures are assumed and labelled so.

2 Basis and assumptions

Item	Basis
C-201 absorbed power	Model: gas power F·cp·ΔT with the compression rise divided by 0.72, i.e. shaft power: 6,528 kW at design (1,928 kNm³/h, 80 % speed); 12,594 kW at 100 % speed (2,415 kNm³/h, copy of the model with FIC-201 in MAN at 100 %, trips off).
C-201 motor rating	≥ 1.10 × maximum absorbed power (API 617 driver margin, typical practice) = 13,853 kW → 14,000 kW (next 0.5 MW step, assumed); 11 kV, variable-speed drive.
Efficiencies, power factors	Motor 0.97, VSD 0.975, VSD input PF 0.95, motor PF 0.9 (assumed); LV motors η 0.90–0.92, PF 0.80–0.85 (assumed).
Demand factors	Continuous 1, intermittent 0.3, standby 0.1 (typical practice, assumed). Maximum demand = Σ input kW × DF; kvar likewise.
UPS	DCS 2.0 kW, SIS 1.5 kW, workstations 1.2 kW, network 1.0 kW + 1.2 W per installed I/O channel; 81 installed I/O from MEOH-V1-IN-INL-001 → 5.8 kW, UPS 10 kVA at ≤ 80 % load (assumed).
Heat tracing	Analyser sample lines only (tropical site, no freeze protection, assumed). AIT-208 maintained at 120 °C: model converter outlet CH3OH 4.78 % at 78.39 bar a condenses below 101.9 °C.
Process pumps	None on MEOH-PID-001/002: crude methanol leaves V-201 under loop pressure through LV-204; cooling water and instrument air are utilities outside this unit.
Lighting	2 400 m² plot (60 m × 40 m, assumed) at 6 W/m² LED (allowance).

3 Load list

No.	Consumer	Tag	Bus	Rated kW	Abs. kW	η	PF	Input kW	kvar	kVA	Duty	DF	Dem. kW	Dem. kvar	Basis
L01	C-201 loop circulator motor via VSD	C-201	11 kV	14,000	6,528	0.946	0.95	6,902.8	2,268.9	7,266.1	C	1	6,902.8	2,268.9	Model: 6528 kW shaft at design (80 % speed), 12594 kW at 100 % speed; rating 1.10 × max absorbed, rounded up to 0.5 MW (typical, API 617 driver margin); motor 97 %, VSD 97.5 %, input PF 0.95 (assumed)
L02	C-201 lube-oil pump A (main)	C-201 aux.	400 V	30	22	0.920	0.85	23.9	14.8	28.1	C	1	23.9	14.8	Assumed (seal/lube system per vendor, MEOH-PID-002 note 4)
L03	C-201 lube-oil pump B (standby, auto-start)	C-201 aux.	400 V	30	22	0.920	0.85	23.9	14.8	28.1	S	0.1	2.4	1.5	Assumed
L04	C-201 lube-oil heater (start-up only)	C-201 aux.	400 V	15	15	1.000	1.00	15.0	0.0	15.0	I	0.3	4.5	0.0	Assumed
L05	C-201 motor anti-condensation heater (motor stopped)	C-201 aux.	400 V	4	4	1.000	1.00	4.0	0.0	4.0	I	0.3	1.2	0.0	Assumed
L06	C-201 VSD cooling and auxiliaries	C-201 aux.	400 V	60	50	1.000	0.85	50.0	31.0	58.8	C	1	50.0	31.0	Assumed (≈ 0.4 % of the drive rating)

No.	Consumer	Tag	Bus	Rated kW	Abs. kW	η	PF	Input kW	kvar	kVA	Duty	DF	Dem. kW	Dem. kvar	Basis
L07	C-201 dry-gas seal gas heater / booster	C-201 aux.	400 V	10	8	1.000	0.95	8.0	2.6	8.4	C	1	8.0	2.6	Assumed
L08	Process pumps	— (none drawn)	—	0	0	1.000	1.00	0.0	0.0	0.0	C	1	0.0	0.0	None on MEOH-PID-001/002: crude methanol leaves V-201 under loop pressure through LV-204; cooling water is a utility (outside the unit)
L09	Analyser shelter: AIT-205 GC and AIT-208 analyser	AIT-205/208	230 V UPS	3	3	1.000	0.90	3.0	1.4	3.3	C	1	3.0	1.4	Assumed 2 × 1.5 kW
L10	Analyser shelter HVAC and purge fans	Shelter	400 V	12	10	1.000	0.85	10.0	6.2	11.8	C	1	10.0	6.2	Assumed
L11	Electric heat tracing: AIT-208 sample line (maintain 120 °C, 30 m × 40 W/m)	AIT-208	230 V	1.2	1.2	1.000	1.00	1.2	0.0	1.2	C	1	1.2	0.0	Assumed length and W/m; maintain temperature from the model outlet dew point 101.9 °C (CH3OH 4.78 % at 78.39 bar a) + margin
L12	Electric heat tracing: AIT-205 sample line (30 m × 20 W/m)	AIT-205	230 V	0.6	0.6	1.000	1.00	0.6	0.0	0.6	C	1	0.6	0.0	Assumed; no freeze protection needed (tropical site, assumed)
L13	Instrument, DCS and SIS UPS (81 installed I/O)	UPS	400 V → UPS	8	5.8	0.920	0.90	6.3	3.0	7.0	C	1	6.3	3.0	Assumed: DCS 2.0 kW, SIS 1.5 kW, workstations 1.2 kW, network 1.0 kW + 1.2 W per installed I/O channel (INL MEOH-V1-IN-INL-001); UPS 10 kVA at ≤ 80 % load
L14	Area and platform lighting (LED), 2 400 m² × 6 W/m²	Lighting	230/400 V	14	14	1.000	0.95	14.4	4.7	15.2	C	1	14.4	4.7	Assumed plot 60 m × 40 m and lighting density (allowance)
L15	Maintenance sockets and welding outlets (2 × 32 kW)	Small power	400 V	64	64	1.000	0.85	64.0	39.7	75.3	I	0.3	19.2	11.9	Allowance, assumed
L16	Catalyst-handling hoist on the R-201 structure	R-201	400 V	15	12	0.900	0.80	13.3	10.0	16.7	I	0.3	4.0	3.0	Assumed (catalyst change only)

Duty: C continuous, I intermittent, S standby. Only L01 absorbed power is from the model.

4 Maximum demand

System	kW	kvar	kVA	Supply (assumed)
11 kV (C-201 drive)	6,903	2,269	7,266	11 kV switchboard feeder to the C-201 VSD
400 V / 230 V auxiliaries, UPS, lighting	148.7	80.2	169.0	11/0.42 kV transformer 250 kVA (≥ 1.25 × demand), u_k 4 % → about 9 kA at the 400 V board
Unit total	7,052	2,349	7,432	overall PF 0.949; C-201 = 97.9 % of the demand

5 C-201 motor

Absorbed shaft power at design 6,528 kW; at 100 % speed 12,594 kW at 2,415 kNm³/h (power rises roughly with the cube of flow: (2415/1928)³ = 1.97 against 1.93 from the model). Rating 1.10 × 12,594 = 13,853 kW → **14,000 kW**. Electrical input at design 6,528 / (0.97 × 0.975) = 6,902 kW. At design the motor is loaded to 47 % of its rating. The indicator JI-201 (0–12 MW) would saturate at 100 % speed: see EQ-ei-05 in MEOH-V1-IN-INL-001.

6 Cable sizing — C-201 motor feeder (worked example)

FLC = P / (√3·V·PF·η) = 14,000 kW / (√3 × 11 kV × 0.9 × 0.97) = **842 A**; design current I_b = 1.25 × FLC = 1052 A (NFPA 70 Art. 430.22/430.122 practice: motor and drive conductors at 125 %, quoted as typical practice).

Installation: 3 runs of 6.35/11 kV 3-core Cu XLPE, armoured, direct in ground at 30 °C, grouped (k_g 0.85, assumed), 150 m (assumed). k_θ = √((90 − 30) / (90 − 20)) = 0.926; k = 0.787; table rating needed per run ≥ 1052 / (3 × 0.787) = 446 A → **3 × 300 mm²** (485 A each, I_z = 1145 A ≥ I_b).

Voltage drop running: ΔV = √3·I·L·(R cos φ + X sin φ) / (n·V) = √3 × 842 A × 0.150 km × (0.0782 × 0.9 + 0.092 × 0.436) / (3 × 11 000 V) = **0.07 %** (limit 5 %, IEC 60364-5-52 Annex G as typical practice). Starting through the VSD (≤ 1.0 × FLC, assumed): cable 0.07 % + board dip I_start/I_k = 3.4 % = **3.4 %** (limit 15 %, assumed). Direct-on-line what-if (6 × FLC, PF 0.2): cable 0.42 %, board dip 20 % — the network, not the cable, limits starting at 11 kV.

Short circuit (adiabatic, IEC 60364-5-54 practice): S ≥ I_k √t / k = 25 kA × √0.25 s / 143 = 87 mm² per run (each run taken to carry the full fault) — 300 mm² satisfies it.

Size mm²	Table A	I_z derated × runs	R90 Ω/km	X Ω/km	ΔV run %	ΔV start %	S ≥ S_min	Meets all
50	180	425	0.4935	0.117	0.33	3.7	no	no
70	220	519	0.3417	0.11	0.24	3.6	no	no
95	265	626	0.2461	0.105	0.18	3.5	yes	no
120	300	708	0.1951	0.102	0.15	3.5	yes	no
150	335	791	0.1613	0.099	0.12	3.5	yes	no
185	380	897	0.1289	0.097	0.10	3.5	yes	no
240	435	1027	0.0981	0.094	0.09	3.5	yes	no
300	485	1145	0.0782	0.092	0.07	3.4	yes	selected

Ratings: typical catalogue values for 3-core 11 kV XLPE cables, direct in ground 20 °C, 1.2 K·m/W, 0.8 m, or in free air 30 °C (assumed). R90 = R20 (IEC 60228 class 2 copper) × 1.275 (+2 % skin effect from 150 mm²).

Second example, C-201 lube-oil pump A (30 kW, 400 V DOL, 7 × FLC start, 120 m on tray at 40 °C, grouping 0.80, board 9 kA, clearing 0.02 s, all assumed): FLC 55.4 A, I_b 69.2 A → **16 mm²** 0.6/1 kV multicore Cu XLPE, armoured (I_z 73 A; ΔV running 3.71 %, at start 14.7 %; S_min 8.9 mm²).

7 Engineering queries

No.	To	Issue	Proposed
EQ-ee-01	MEOH-PID-002 note 4 (C-201 vendor)	C-201 seal and lube-oil systems are "per vendor": the lube-oil pumps, heaters, seal-gas and VSD auxiliary loads in the load list are assumed.	Obtain the C-201 package auxiliary load list; show the lube-oil console and the seal-gas panel on the P&ID.
EQ-ee-03	MEOH-PID-001	The AIT-208 sample (converter outlet, CH3OH 4.78 % at 78.39 bar a) condenses below about 101.9 °C; no heat tracing is shown on the analyser sample lines.	Show electric heat tracing on the AIT-205/208 sample lines (maintain ≥ 120 °C on AIT-208) on MEOH-PID-001 rev B; keep the load in the load list.
EQ-ee-05	Simulator model (C-201 curve) / MEOH-PID-002	The motor rating (14 MW) is set by the model's 100 % speed point (12.59 MW absorbed at 2415 kNm³/h); the model has no maximum continuous speed or surge limit and the P&ID gives no driver rating.	Confirm the C-201 maximum continuous speed and the rated point with the compressor curve; show the driver rating on MEOH-PID-002.

Queries raised against the hazardous-area classification are listed in MEOH-V1-EE-HAC-001; the instrument-list queries (e.g. EQ-ei-05, JI-201 range) in MEOH-V1-IN-INL-001.